ABHIMANYU (RAJ) SINGH *CHANDEL*

Scan to know more about the Project.

PROFILE

Motivated environmental scientist passionate about biodiversity and ecological problem-solving. Committed to applying innovative solutions, I aim to support evidence-based decision-making and applied conservation outcomes through work in ecological modelling, terrestrial habitat assessment, and invasive species control. My goal is to leverage my skills as an Environmental Scientist or Spatial Ecologist to apply emerging technologies into biodiversity conservation.

SKILLS – Office and Field Science

Research Design/ Plan
Teaching/Training/Writing
Species Id Skills/Spotting
inter-intra species interactions
Habitat Suitability/Resilience
Field work/data collection
Manual Driving/4WD/Machinery

Technical

Ecology Cloud Software testing
Automation QA testing (Selenium/playwright)
Bug fixing /dev notebook
Quarto/rmd/.ipynb code
Apple/Microsoft Ecosystems
Google Earth/Geodatabases
Spatial data analysis (GIS/R/py)

Admin

Problem Solving
Management/JIRA/Slack
Good Communication Skills
Customer accountability and
Stakeholder Management Team
Conceptualist and Collaborator

Art & Fun

Bird Watching
Singer-songwriter
Explorer/Climber/Hiker
Photography/ Graphic designing

RECENT WORK EXPERIENCE

2024 JAN - July 2025, **QCIF Sustainable Futures**, QLD-NSW**,** EcoCommons/BCCVL

- **Data Ecologist (Full time)**
- **Communication Officer & User Support (Full time)**

2021 NOV - DEC 2023, **Griffith University**

Dept. **eResearch** & Specialised Advisory, Research and Data Foundations Brisbane, QLD. EcoCommons/BCCVL

- **Functional Ecology Analyst & User Support Training Officer (Full time)**
- **Functional Ecology Analyst (Casual)**
- **Student Data Ecology Analyst (Intern)**

Studying species requirements and ecological interactions and developing use cases for **species of concern** tailored to government, research, and conference contexts. Conducting climate scenario testing, **projections**, and ecological modelling using environmental datasets from multiple sources and tools like R, Python, and Jupyter, incorporating **models** such as CMIP6 and **scenarios** such as SSP2-5 and RCPs for future projections, while adhering to FAIR data principles and applying metadata standards for dataset integration and reproducibility. Developing **workflows** and test cases; testing features, resolving spatial errors, and supporting technical development, including contributions to **reproducible**, **open-access ecological workflows** and national-scale data integration through my work with EcoCommons. **Science communication**, delivering user training and technical support through a range of formats including **workshops**, online **tutorials**, and live **demonstrations**. This involved helping users apply **species distribution modelling (SDM)** tools within the platform and working closely with stakeholders across research, government, and environmental consulting sectors to support applied conservation and planning needs. **Supervising** research students, mentoring colleagues, and supporting administrative functions including AGILE sprint planning. Bridging information and research gaps between **technical and scientific teams** by aligning modelling outputs with research needs and coordinating across multidisciplinary stakeholders to ensure shared understanding and effective tool integration.

2021 JAN – FEB 2022, **BUSH-IT** Sydney, NSW

- **Environmental Field Officer/ Bush Regenerator (Casual)**

Ecological restoration across highly degraded sites (e.g., quarries, landfills). Invasive flora management (e.g., Madeira Vine, Ochna), **assisted** natural regeneration with **endemic** species, and **biodiversity asset maintenance**. Applied field machinery and techniques like scrape-and-paint, tuber removal, and **site-based** monitoring. **Pre-burn** preparation, **WSUD** (Water Sensitive Urban Design), stormwater management, and native species enrichment. **Remote** bushland work with natural area **protection** focus.

2020 AUG – APR 2021, **SITE ENVIRONMENTAL AND REMEDIATION SERVICES (SERS)** Sydney, NSW

- **Environment Field Technician (Casual)**

Environmental monitoring field systems **setup** and **calibration**. I drove to various **remote** sites in NSW regarding installation, maintaining and monitoring equipments, accounting environment pollution and disturbance, generating data for **council** reports and **construction** companies.

SCIENTIFIC PUBLICATION

2025 Jan

- EcoCommons Australia virtual laboratories with cloud computing: Meeting the needs of ecological and environmental research. (2025). ***Environmental Modelling & Software***, 105905.

PROJECT HIGHLIGHTS

Australia

Use cases for ACT/QLD/NSW Government
Science Workshops/ Conferences/Symposium
Invasive species management, Control burn prep and Assisted native reforestation for NSW councils and land managers.
Seed Collection, Soil-sapling for Lane Cove National Park Nursery
Climate Dividend Awareness City of Ryde
Sea bin Project/ Collecting micro-plastic data
Moreton island micro plastic collection

USA

Monarch Butterfly Migration Tagging
Horseshoe crab surveys

India

Seed Ball Program/ Urban conflict Tiger survey with MP Biodiversity Council
Tiger and Vulture Census/ Reptile Workshop/ Plantation Drives/Tiger Stamp Painting Awareness/Wildlife Quiz
Science Arts and Fun Mobile Education Van

CONTACT

Phone – 0492914270
Location- Sydney (Ryde)
Email –abhirsc@outlook.com
Social - @naturality_abhirsc

LinkedIn:

FIELD OF INTEREST

#Terrestrial_Ecology #Invasive_species #Zoology #Insect_conservation #Pytoremediation #Land_managemnet #Climate_Change #Ecological_Restoration_Technologies #Adaptation#Mitigation#Climate_change_ready_vegetation #EIA #PSI #BAM #BDAR #CPR #LCA #ESG #SDG #CSR #EHS #UAV #ENVI #LiDAR #Naturalist#Bioacoustics

VOLUNTEER WORK EXPERIENCE

2022 JUNE – SEPT 2023, **Green Infrastructure and Research Labs (GIRLS),** Sydney, NSW

- **Ecology *volunteer***

Assisting PhD students designing experiments and ecological discussion, awareness interactions events, connecting with latest resources and technology.

2021 JUNE – DEC 2021, **SEA BIN**, Sydney, NSW

- **Micro-plastic *volunteer***

Sorting sea bin collected waste and reporting Micro-plastic, Styrofoam's, etc.

2021 APRIL – DEC 2021, **CLIMATE CITIZEN'S LOBBY**, Sydney, NSW

- **Event *volunteer***

Community engagement volunteering, Tabling events, Climate dividend.

2020 JUNE – DEC 2021, **BUSH CARE, PARAMATTA COUNCIL**, Sydney, NSW

- **Bushcare *volunteer***

Bush-care regeneration volunteer at Stanley Bush-care, Parramatta Council.

2020 FEB - APR 2020, **UPPER MOUNT GRAVATT BUSHCARE GROUP**, Brisbane City Council, QLD.

- **Bushcare *volunteer***

To learn and identify local biodiversity. Picking invasive weed plants and litter and support native flora and fauna.

2020 FEB, **CLEAN UP AUSTRALIA LTD.**, QLD

- **Student *volunteer***

Picking invasive weed plants and litter from beaches and hiking trails to support native flora and fauna.

PREVIOUS WORK EXPERIENCE

2020 NOV - 2021 JULY, **GARDENING4U**, Sydney, NSW

- **Gardner (Casual)**

Towing and machinery experience, Backyard, and lawn maintenances. Planting, designing, maintaining, hedging, lawn, pruning and weeding.

EDUCATION

2020 – 22, **GRIFFITH UNIVERSITY**, Brisbane, QLD

MS Environment Science (Environment Protection and Climate Change Adaptation)

- **Environment Management Systems**, Economics, Pollution, Policymaking and Planning Climate Change Mitigation and Adaptations, Land Remediation and Rehabilitation, **Conservation Research** Design and Planning, **Ecological Modelling and Data Management as well as GIS and Remote Sensing**. Air Quality Monitoring and Control, Water and Wastewater treatment, Hazardous waste, and Solid waste management.

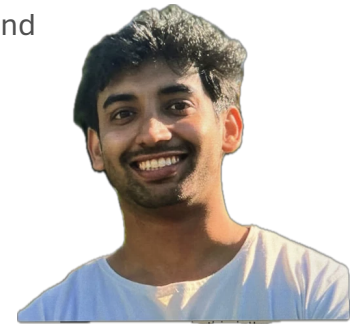
2015- 18, **IGNOU UNIVERSITY**, Delhi, Bhopal, India

Bachelor of Life Science – BS Zoology/Animal Biology

- Taxonomy, Evolution, Diversity, **Ecology**, Human Environment, Nutrition, Genetics, Cell Biology, Atoms and Molecules, Biochemistry, Developmental Biology, **Animal Biology**.

Memberships 2025

- The Environment Institute of Australia and New Zealand **(EIANZ)**
- Ecological Society of Australia **(ESA)**



☺